



ELECTRONICS CLASS (/class/Electronics-Class/)

LESSON 6: CAPACITORS

with randofo

Some of the earliest capacitors were simply glass jars filled with salt water and wrapped in metal foil. These capacitors - called Leyden Jars - were crude devices which stored high voltage electric charge. They helped early experimenters gain a grasp on electricity and how to harness it. Over the years the science behind capacitors has obviously become more refined, and jars have been phased out. Capacitors can fit onto the smallest of circuit boards and hold varying amounts of charge. In this class we will learn about capacitors and make a small vibrobot (a robot which moves using vibration).



What Is a Capacitor?

To put it simply, a capacitor is a component that stores electricity and then discharges it back into the circuit when there is a

drop in voltage.

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It is important to stress that capacitors do not equal batteries.

The difference between a battery and a capacitor is that in a battery power is generated through a chemical reaction, and in a capacitor a charge is stored and maintained in an electrical field. A battery can yield much more energy for a longer period than a capacitor. On the other hand, a capacitor - even with much less power storage - can discharge considerably more energy exponentially faster than a battery. This is ideal when you need a lot of energy fast.

◦

A capacitor consists of conductive metal plates separated by an insulating material called a dielectric. The dielectric can be anything such as ceramics, plastics, oxidizing metals, glass, and paper. For instance, if you crack open an electrolytic capacitor you will find it's just two metal plates wrapped in a chemically coated film. When electricity is applied, the metal conductors polarize the electrons in the film and store an electric field.

◦

If you were to look at this old-fashioned adjustable capacitor you will notice that it is just a series of metal plates that slide between one another without touching. In this scenario, the air is the dielectric. If it seems weird that air can be charged to hold an electrical field, think of lightning and the Jacob's Ladder project.

◦

Depending on how much physical overlap there is between plates determines how large of an electrical field can be held. The more surface area that the two sets of plates shares between them, the greater the electrical field will be, and the more capacitance it will have.

◦

On account of their unique ability to store charge and maintain an electrical field, capacitors tend to try to resist changes in voltage within a circuit. As such, capacitors are often connected between power and ground to maintain a steady power supply and filter out intermittent voltage spikes and drops.

◦

Capacitors are measured in Farads, which simply is symbolized with a capital F. Although, keep in mind that the values that you will typically encounter are in the picofarad (pF), nanofarad (nF) or microfarad (uF) ranges.

◦

The two schematic symbols associated with capacitors are pictured above. The less complex one on the left is for non-polarized capacitors and the more complex one on the right is for polarized electrolytic capacitors.



Capacitors can be either polarized or non-polarized. Put simply, a polarized capacitor has a positive lead which needs to be connected to power, and a negative lead which needs to get connected to ground. Nonpolarized capacitors can be connected any which way.

Ceramic disc and mylar film capacitors are non-polarized.

Electrolytic capacitors are polarized.

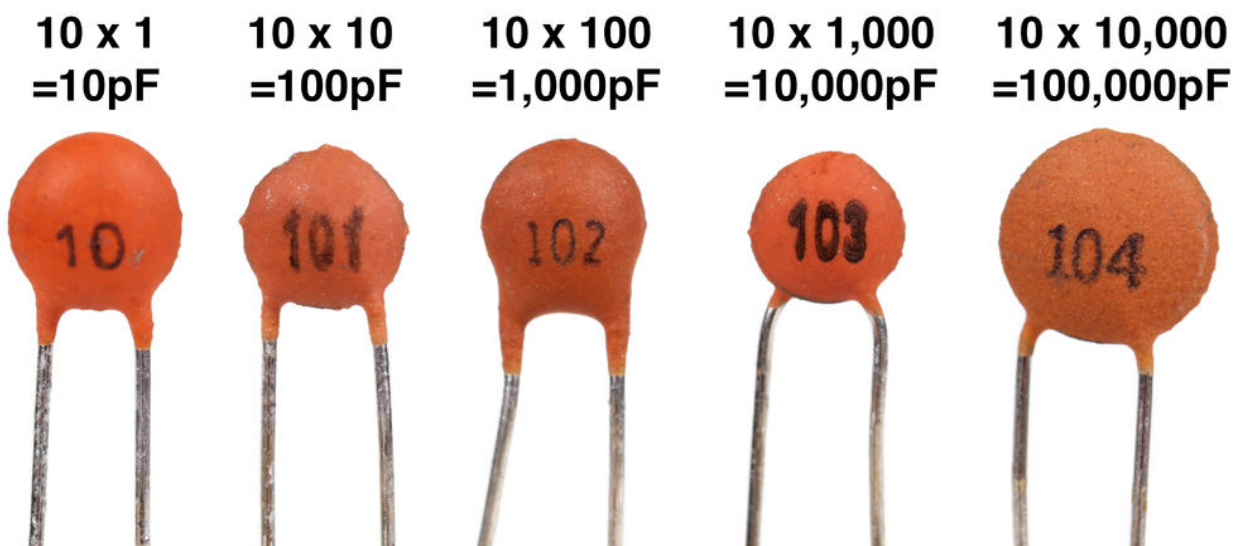
NONPOLARIZED

POLARIZED



Just like resistors, capacitors have markings that need to be deciphered.

Ceramic disc capacitors have two to three digits printed upon them. The first two numbers describe the value of the capacitor and the third number (should it exist) is the number of zeros in the multiplier. When the first two numbers are multiplied against the multiplier, the resulting solution is the value of the capacitor in picofarads. If there is no multiplier, you just read the value of the first two numbers in picofarads.



For instance, if a capacitor says 10 upon it, it has a multiplier of 1 and is simply read as 10pF. On the other hand, if it followed by a 1, it gets multiplied by 10 to become 100pF. Each subsequent number shifts the multiplier over one decimal point. By the time you get to 104, 10 is being multiplied by 10,000.

10pF 0.01nF 0.00001uF	100pF 0.1nF 0.0001uF	1,000pF 1nF 0.001uF	10,000pF 10nF 0.01uF	100,000pF 100nF 0.1uF
				

The one tricky bit about capacitors is that they can be described in picofarads (pF), nanofarads (nF) and microfarads (uF). The measurement you use is determined by whatever makes the most sense to describe the value. You would not describe a 10pF capacitor as a 0.00001uF capacitor any more than you would describe a 0.1uF capacitor as a 100,000pF capacitor. As a general rule, any number that requires more than two zeros to express it is excessive.

Reading electrolytic capacitors is a bit easier. You just literally read them. They have their capacitance and voltage rating printed directly upon them. We have yet to touch upon voltage rating. Suffice it to say that the voltage rating is only important in that your supply voltage should never exceed it. This is unlikely for the type of electronics we are doing.

On electrolytic capacitors, the most important thing to pay careful attention to is the stripe marked with a minus sign indicating the capacitor's negative lead.

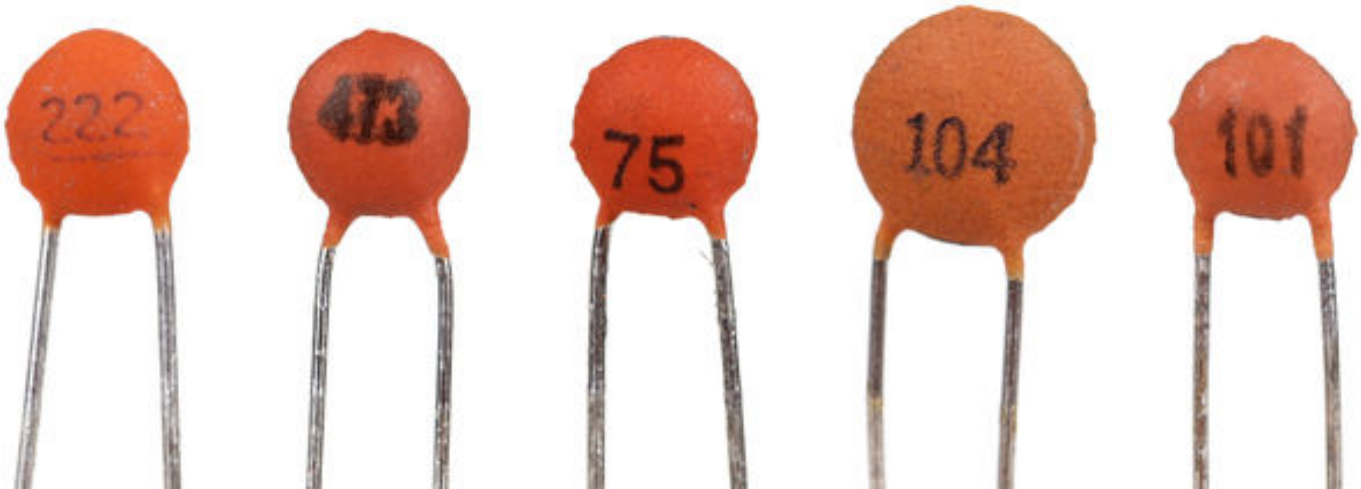
2.2nF

47nF

75pF

0.1uF

100pF



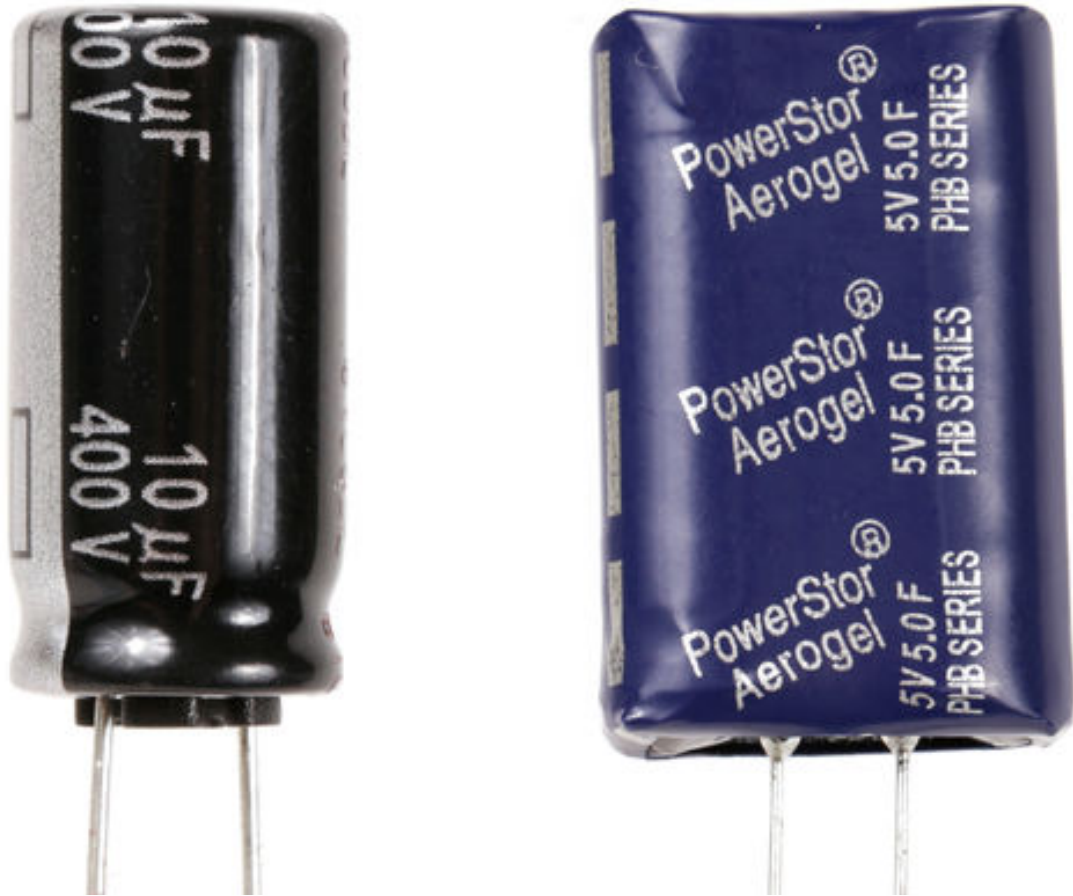
High Voltage Vs. Supercapacitor

Capacitors typically are rated for a small amount of charge and voltage. However, there are two special types of capacitors that buck this trend.

High voltage capacitors - as the name would imply - are capable of storing a large amount of voltage. These are often found in camera flashes and some electronic appliances that plug into the wall, such as old-fashioned tube televisions. You need to be careful when encountering these because they can maintain charge for quite some time after being charged and will shock the heck out of you if you touch the leads.

Supercapacitors on the other hand typically are very low voltage, but store a heckuva lot of charge. For instance, the smallest capacitor you might encounter in working with electronics is 1pF. If we were to convert this unit to Farads, it would be equivalent to 0.000000000001F. In short, a Farad is really - really - big. Thus, the 15F supercapacitor we are going to use in the vibrobot project stores a deceptively large amount of energy.

While you can touch the leads of a supercapacitor without worrying about electric shock, if you were to accidentally cross the leads of a supercapacitor with something conductive, it could rapidly release enough energy to melt metal. You may not get shocked, but if you are careless, you could get badly burned.



Supercapacitor Vibrobot

For this project we are going to take advantage of supercapacitors to power a vibrobot. In other words, we are going to be using 15F capacitors to power vibrating motors to make robots which move around through vibrations. The basic model has an on/off switch and a charging port to allow it to be charged up between uses. The more advanced version also includes a small solar cell to let it be charged by the sun when not being used.

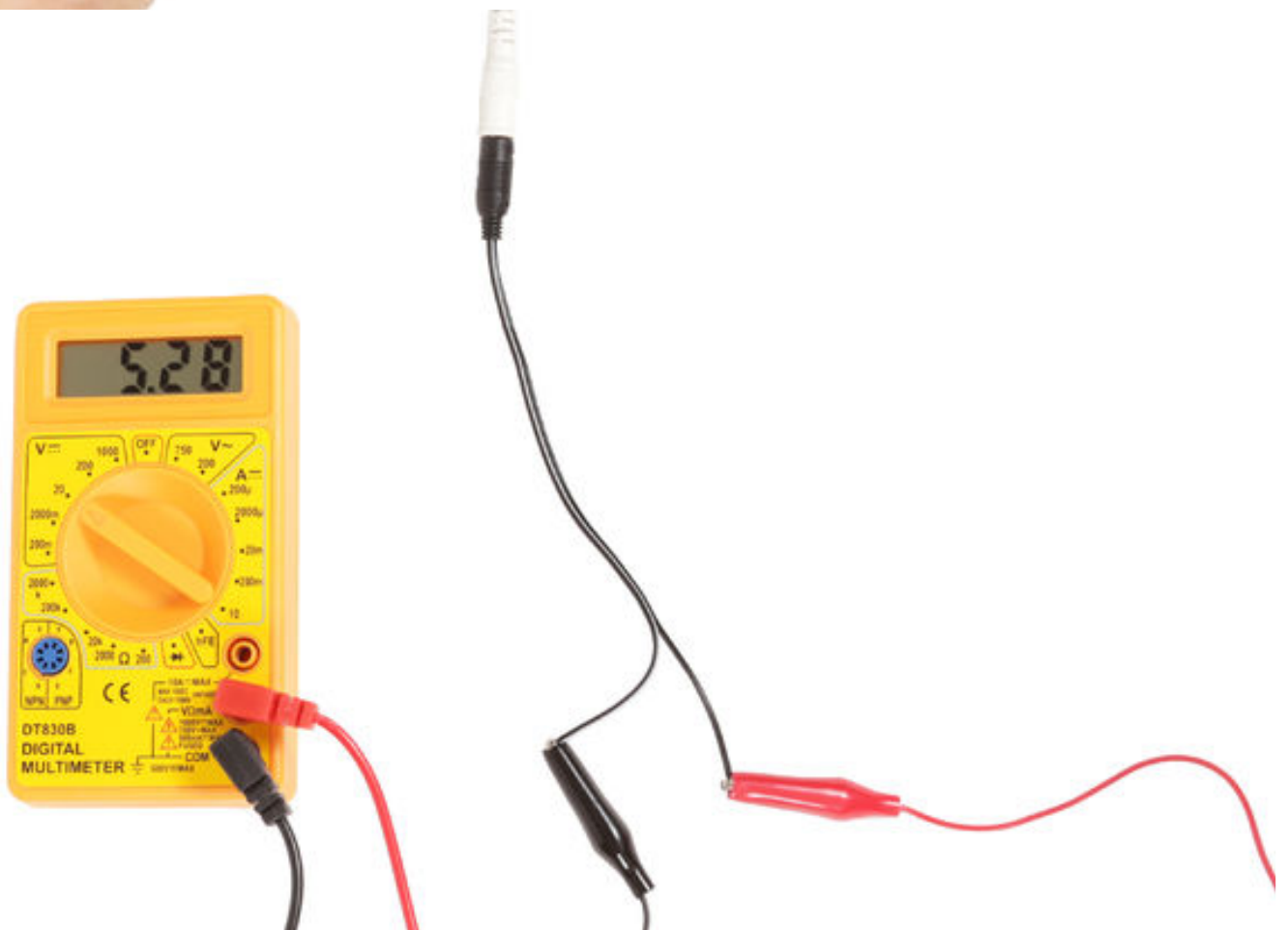


Determine the Polarity

We are going to use a 'wall wart' AC to DC converter to charge the vibrobot.

In order to do this, we need to first determine the polarity of the plug connected to the wall wart to determine which end is positive and which is ground.

Plug the 2-wire adapter into the socket at the end of the cable. Use the voltage setting on your multimeter to measure the voltage coming off of the adapter. If you see a positive voltage, then the wire connected to the red probe is positive and the wire connected to the black probe is ground. Mark these wires to tell them apart if they are not marked already.



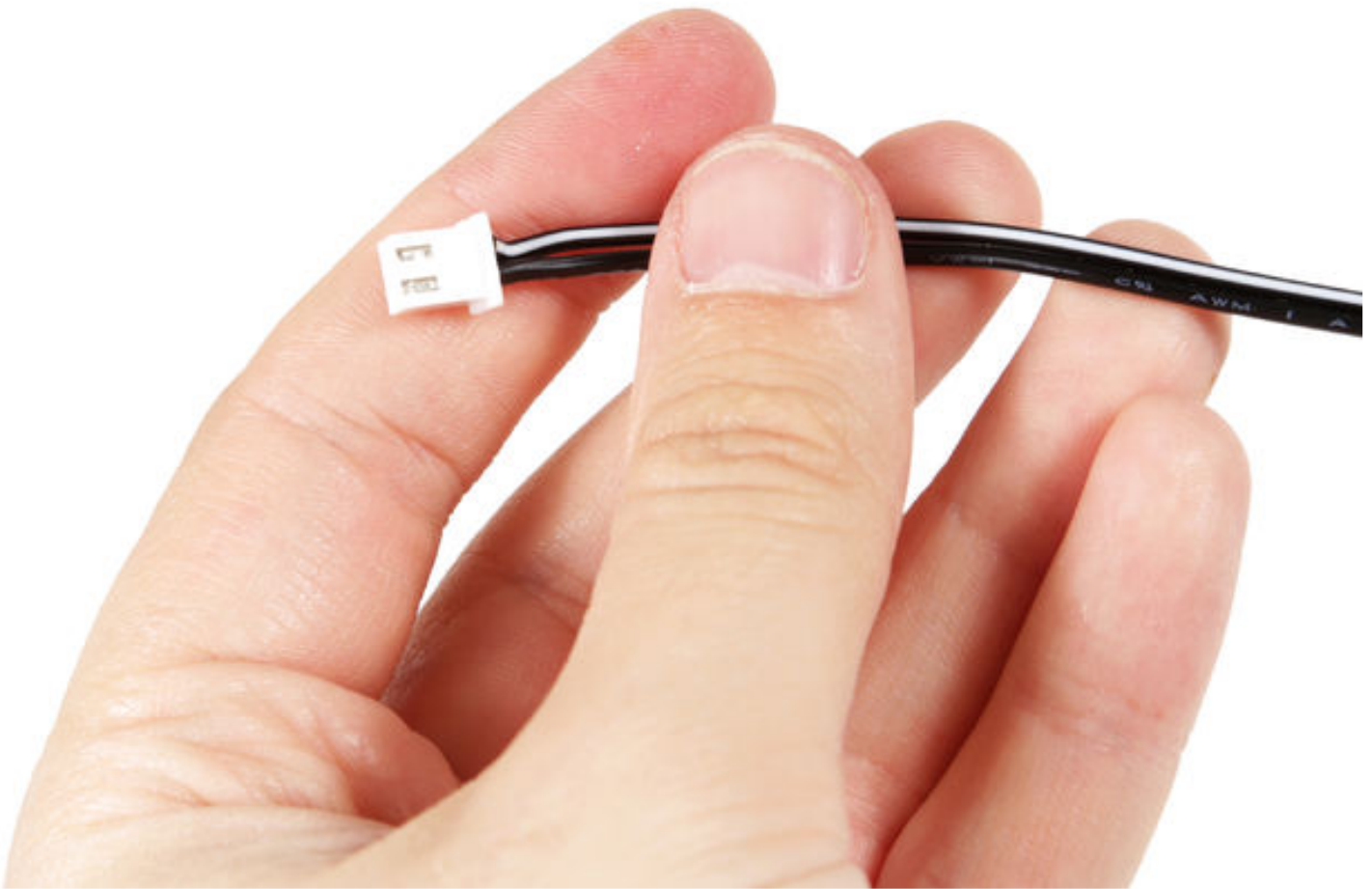
Connector

Solder the metal sockets for the 2-pin female connector onto the end of each wire of the 2-wire power adapter.

Make note of the alignment tab on the plug. If the alignment tab is facing you and the connector is pointing up, ground should be on the left and power should be on the right.

Compress the the metal tabs on the end of each pin and then insert both into the proper socket of the plug by pressing them firmly.

If you are not confident, you can plug the power adapter in and measure with the multimeter to make sure you got it right.









The Circuit

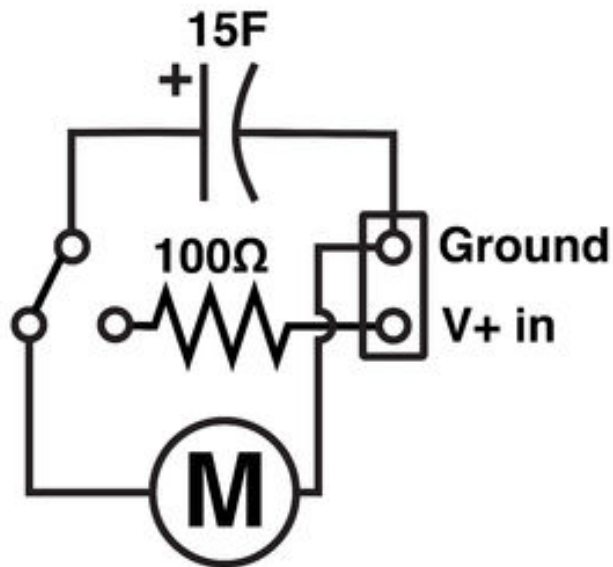
The vibrobot circuit is fairly straight-forward.

There is the charging power that has a power and ground connection.

Ground gets connected to the capacitor and the motor.

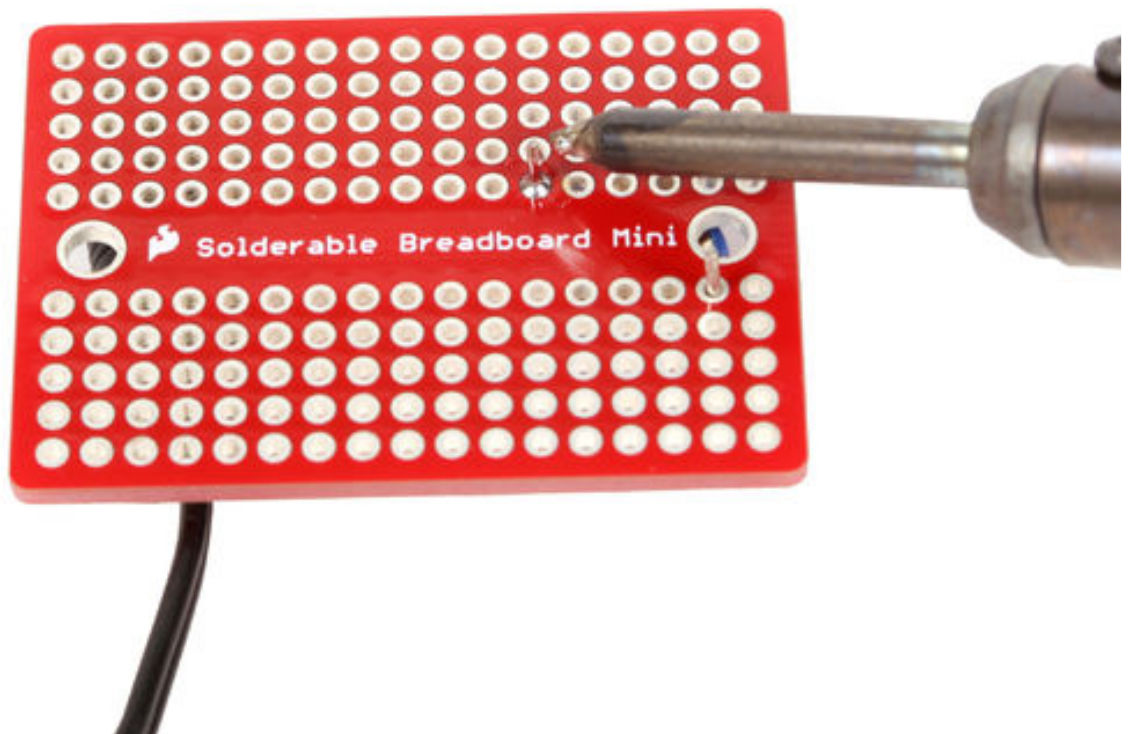
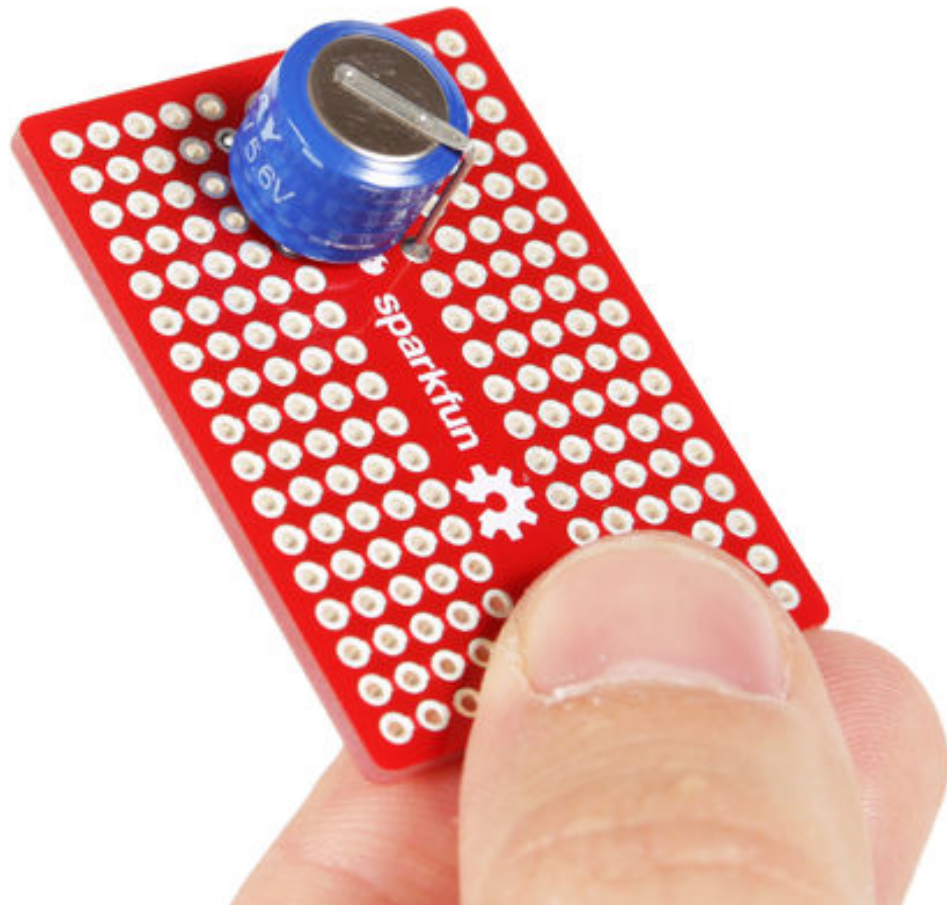
The power input goes to a SPDT switch through a 100 ohm current limiting resistor.

The SPFT switch toggles the capacitor's positive connection between the charger and the motor. In this way, it allows the capacitor to either be charged by the input port or power the motor.



Attach the Capacitor

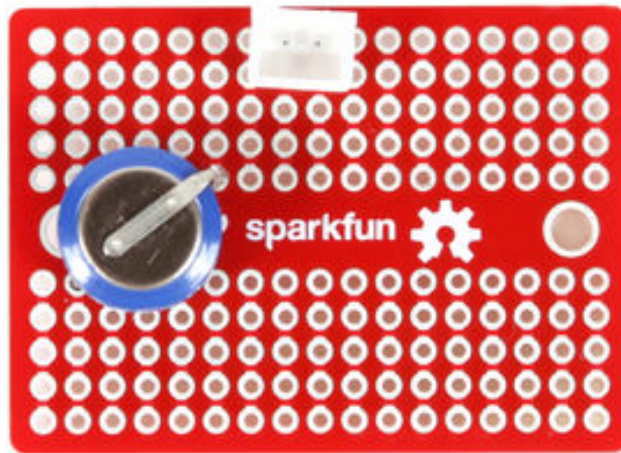
Let's start the circuit board by soldering the supercapacitor in place. Notice that the capacitor has a metal plate on the bottom connected to the power pin. You need to be particularly careful not to accidentally short the power by having the bottom of the capacitor touch any bus rows on the circuit board that might be connected to ground. To easily prevent this, I installed my capacitor at a 45 degree angle straddling the center of the board. This arrangement ensures that a short between power and ground like this likely won't happen.

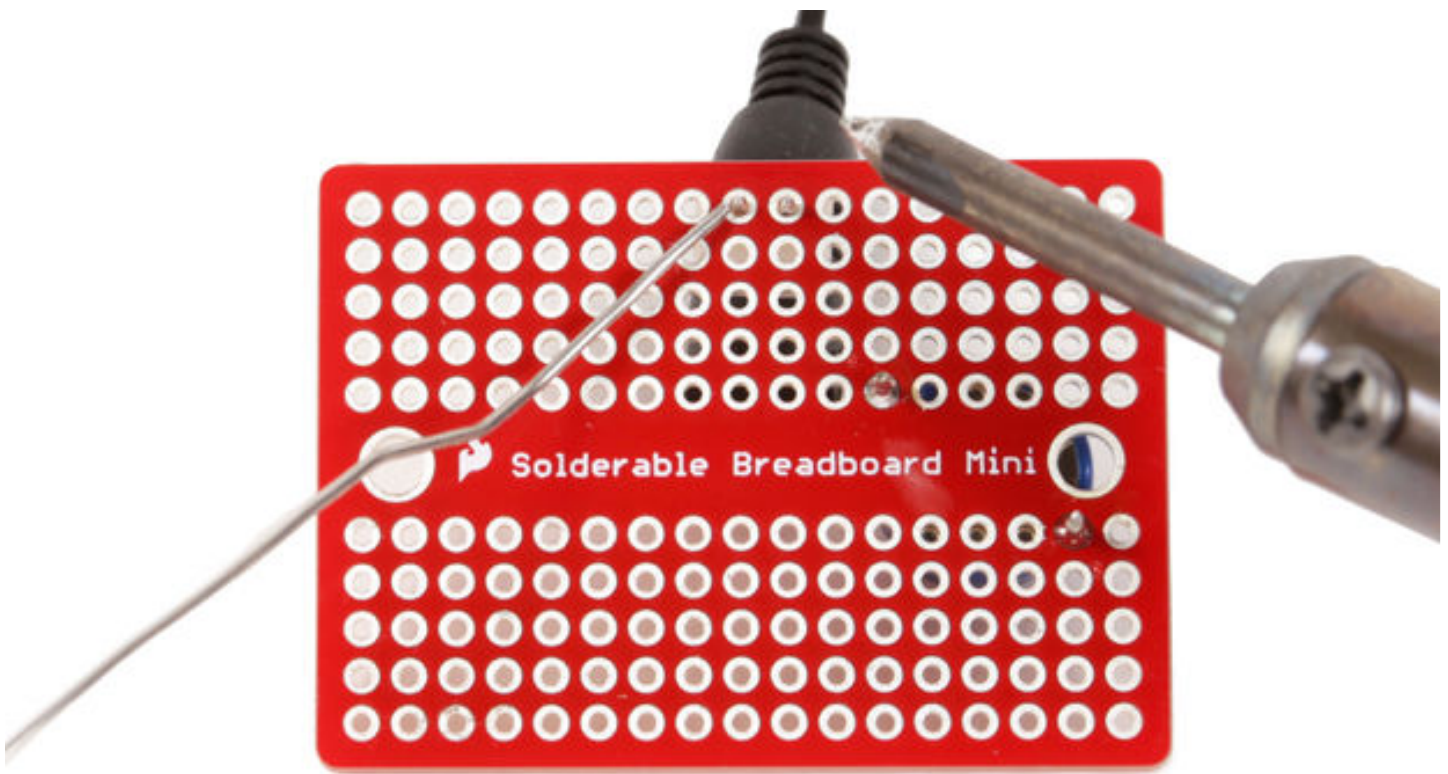


Install the Socket

The next thing to install is the female socket for the power plug. Place this on the same side of the board as the capacitor's ground lead. Place it somewhere in the middle with the indent for the plug's tab facing outwards away from the board.

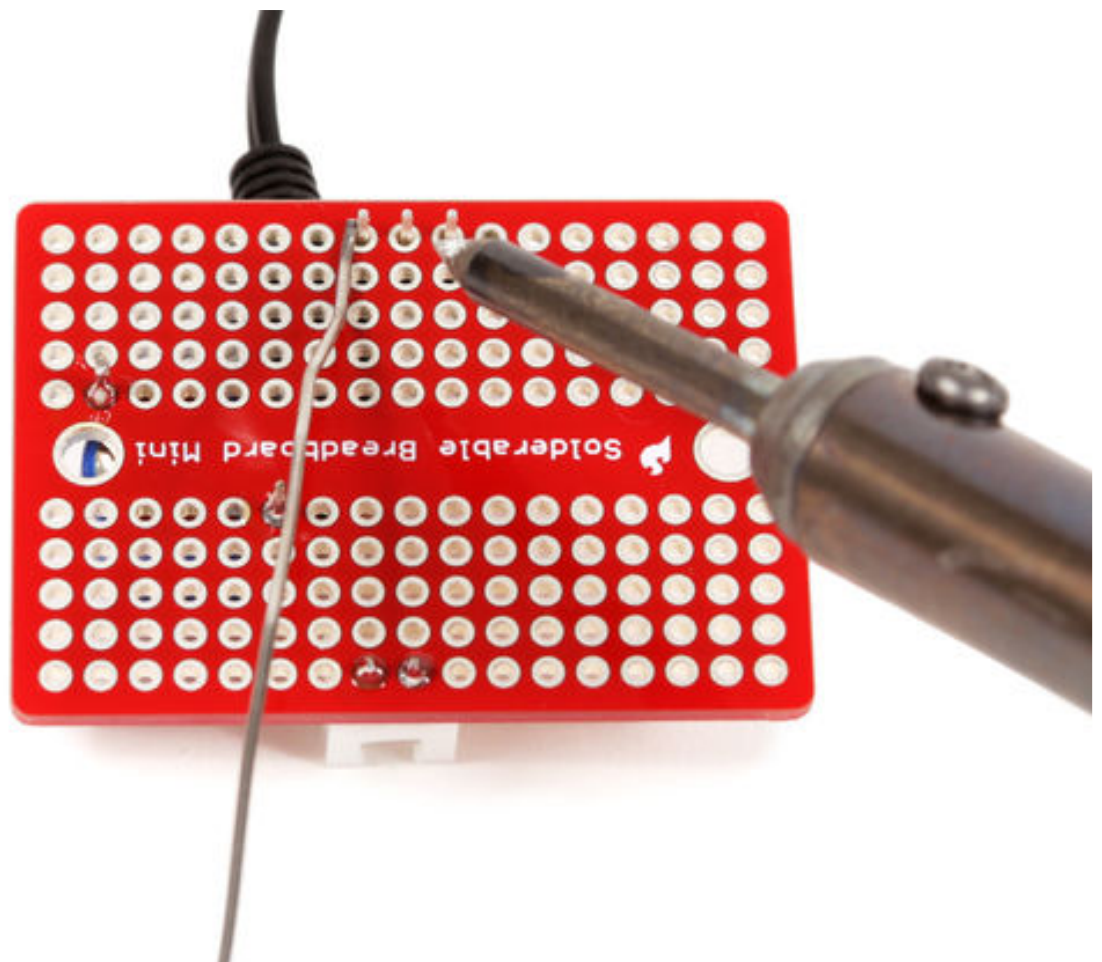
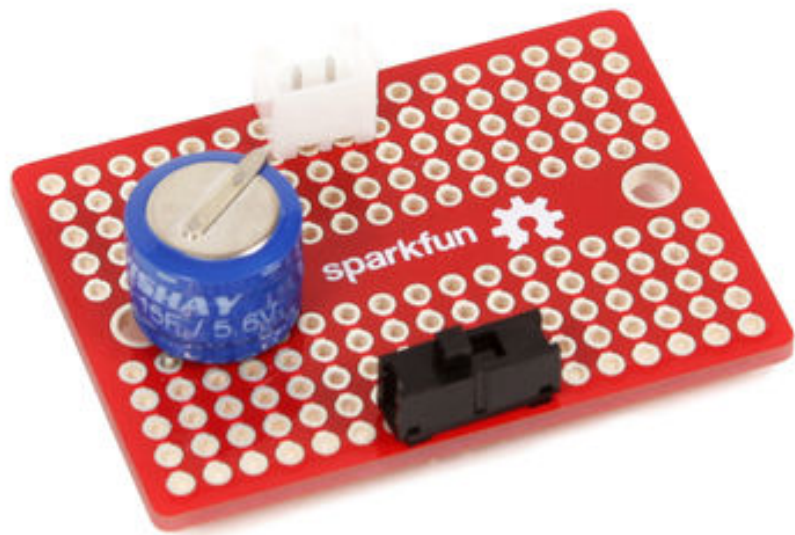
Note that I have something jammed under the board in the soldering picture. This is to hold the component in place while I solder it.





Switch

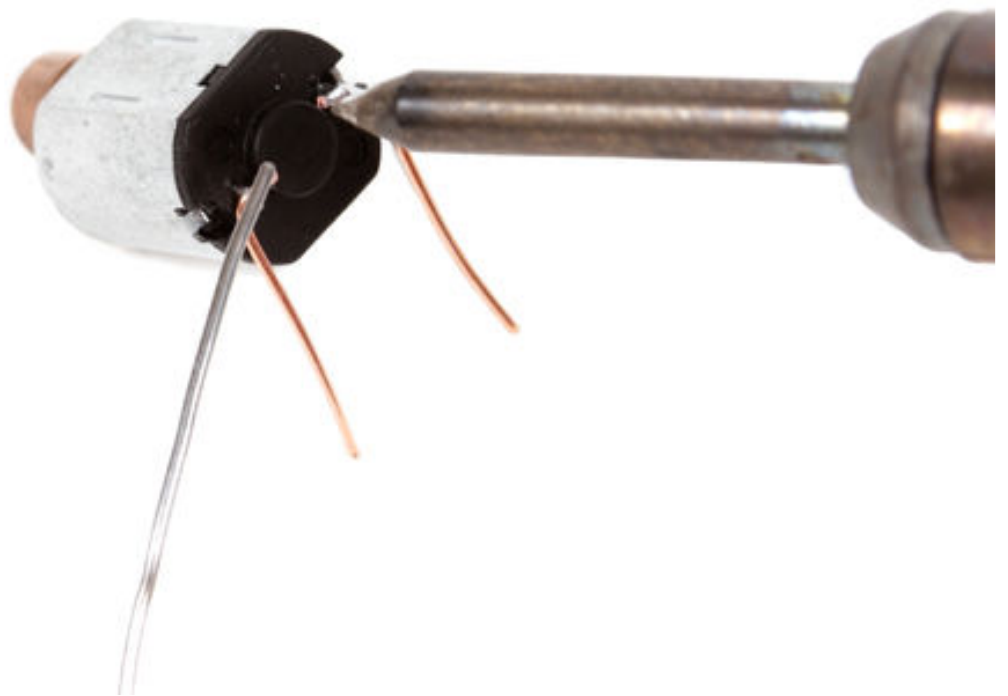
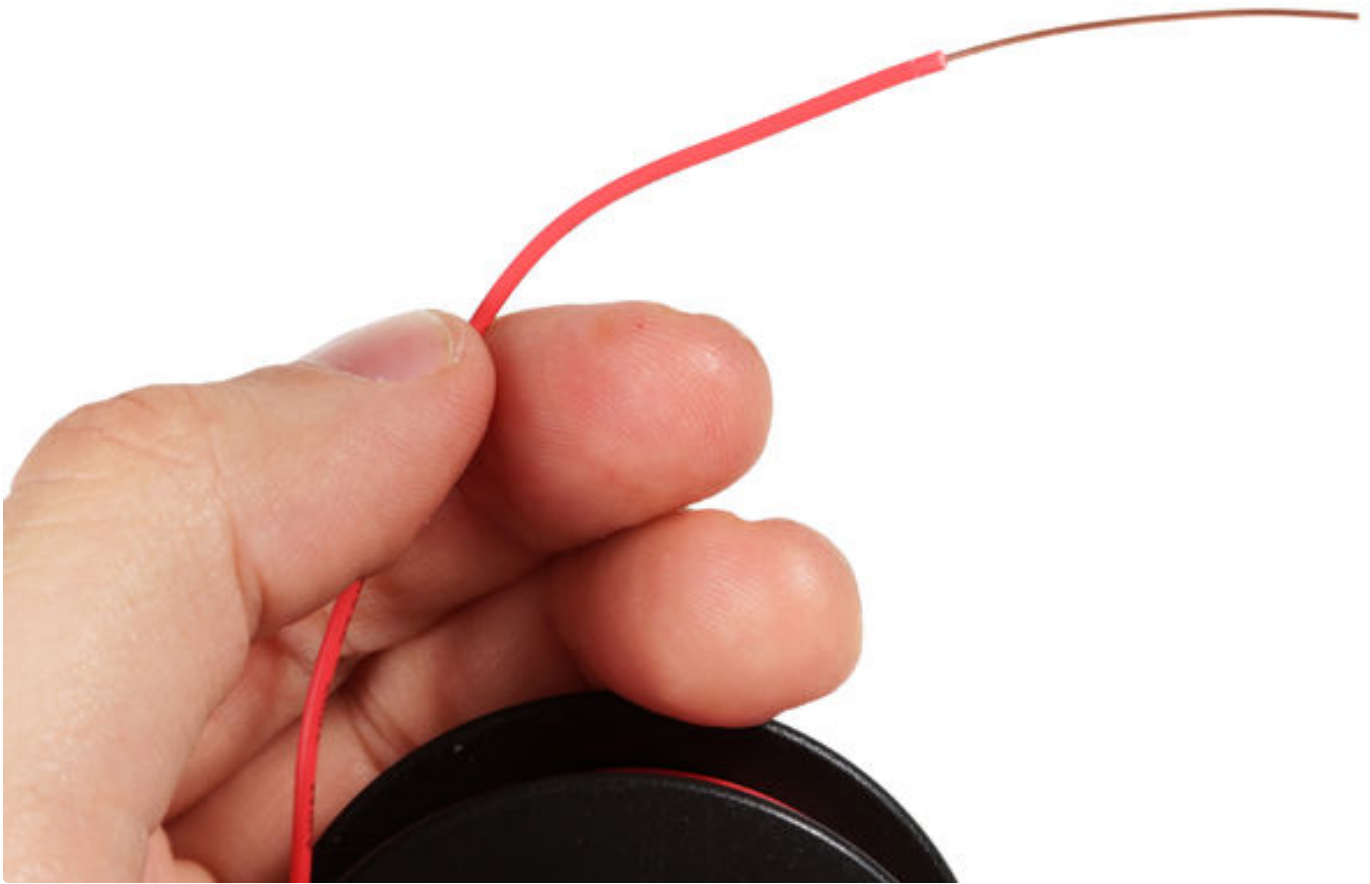
Install the on/off switch on the side of the board opposite from the charger socket.



Wires

Strip about an inch of insulation off the end of solid core wire. Attach the uninsulated wire to one of the terminals on the vibrating motor. Repeat this process for the other terminal.

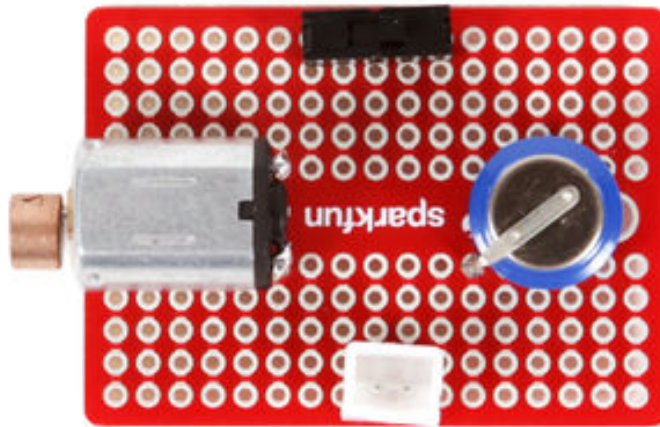


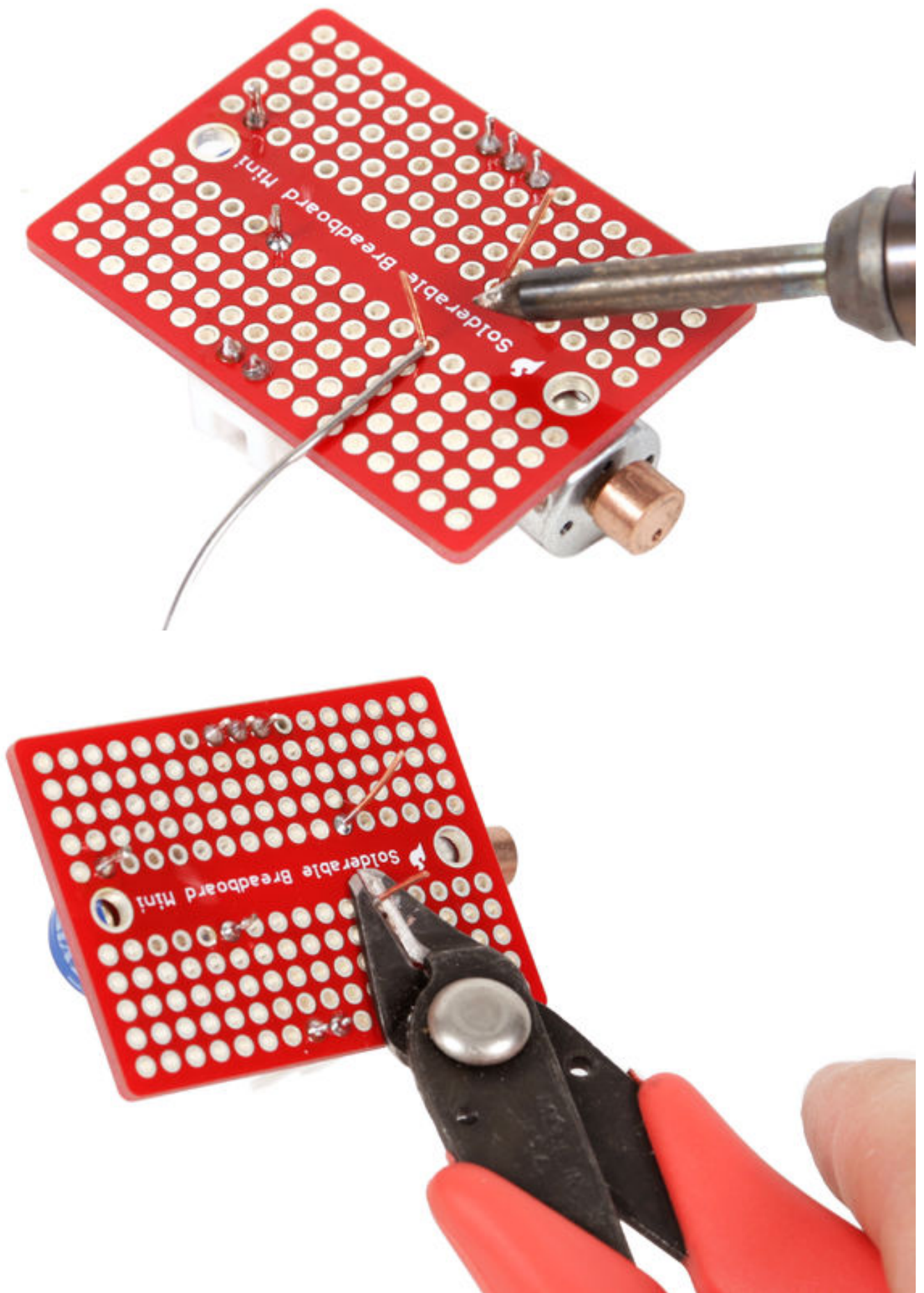


Wire in the Motor

Place the motor centered upon the edge of the board such that its counterweight hangs over the edge.

Insert each motor wire through one of the sockets on their respective sides of the circuit board, and solder them in place.





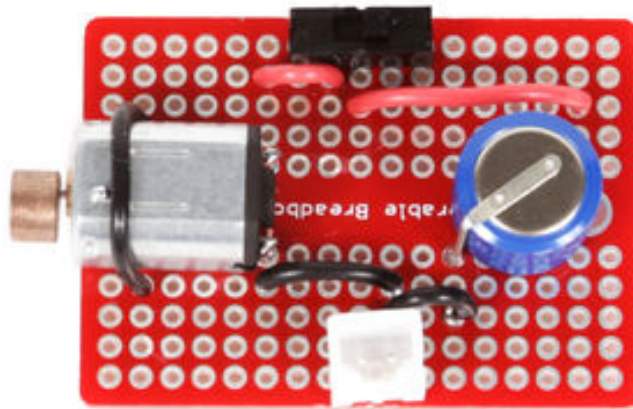
More Wiring

Attach black ground wires between the 2-pin female socket, the ground pin on the capacitor and one of the motor pins.

It is critical to get the connection between the ground pin on the socket and the supercapacitor correct. If you were to reverse it and charge the capacitor backwards, very bad things can happen. So... double check this and make sure you are getting it correct. When the plug is inserted, the ground pin should be wired to the pin with the negative marking on the capacitor.

Once you are absolutely sure you got the ground connections right, solder a red wire between the center pin on the switch and the positive pin on the capacitor. Also solder a red wire between one of the outer pins on the switch and the motor.

Finally, solder a wire around the body of the motor. This shouldn't be electrically connected to anything. It just holds the motor in place.



Charging Resistor

Solder a 100 ohm resistor between the voltage pin on the power socket and the unused pin on the switch.

This resistor is used for charging. If we didn't use the resistor, the supercapacitor will try to draw as much current as it possibly can from the charger. This sudden surge will essentially be like a short wire and possibly either damage it, or if it has protection circuitry, do nothing at all.

The resistor we are using was calculated using Ohm's Law. To be on the safe side, I upped the value slightly since resistors are not perfect, and it can't hurt to have a little bit more.

All of that said, the particular supercapacitor that is being used here has a relatively high internal resistance. What this means is that it does not draw power from a charge as fast as a normal supercapacitor. In fact, it takes an exceptionally long time to charge (about an hour as opposed to 10 seconds).

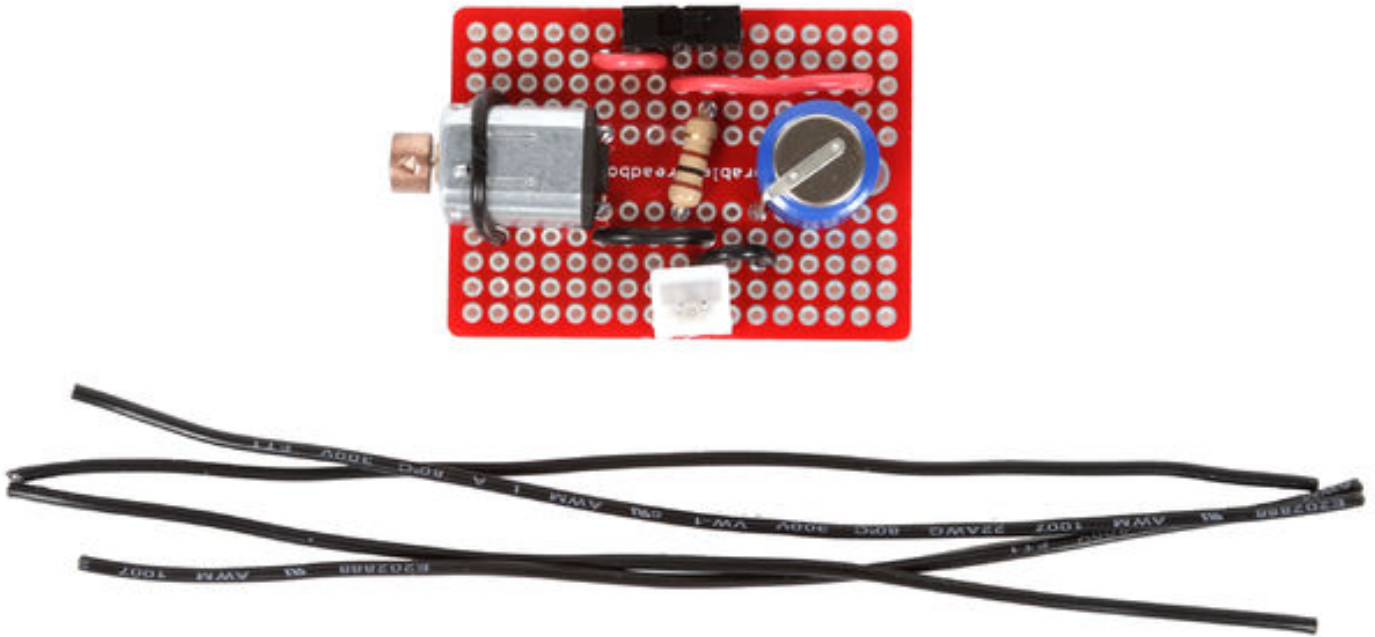
The resistor we are using may not be necessary and might actually slow down charging times a bit. Nevertheless, I have included the resistor in case someone decides to use a different supercapacitor.

You may be wondering why I have chosen to use this one if it charges so slowly. Well, it holds 15F of power, and is a fraction the size of normal supercapacitors. Basically, this little cap holds 3X more power than a supercapacitor that is 5X the size. It may take a while to charge, but it can run for a relatively long time.



Cut Wires

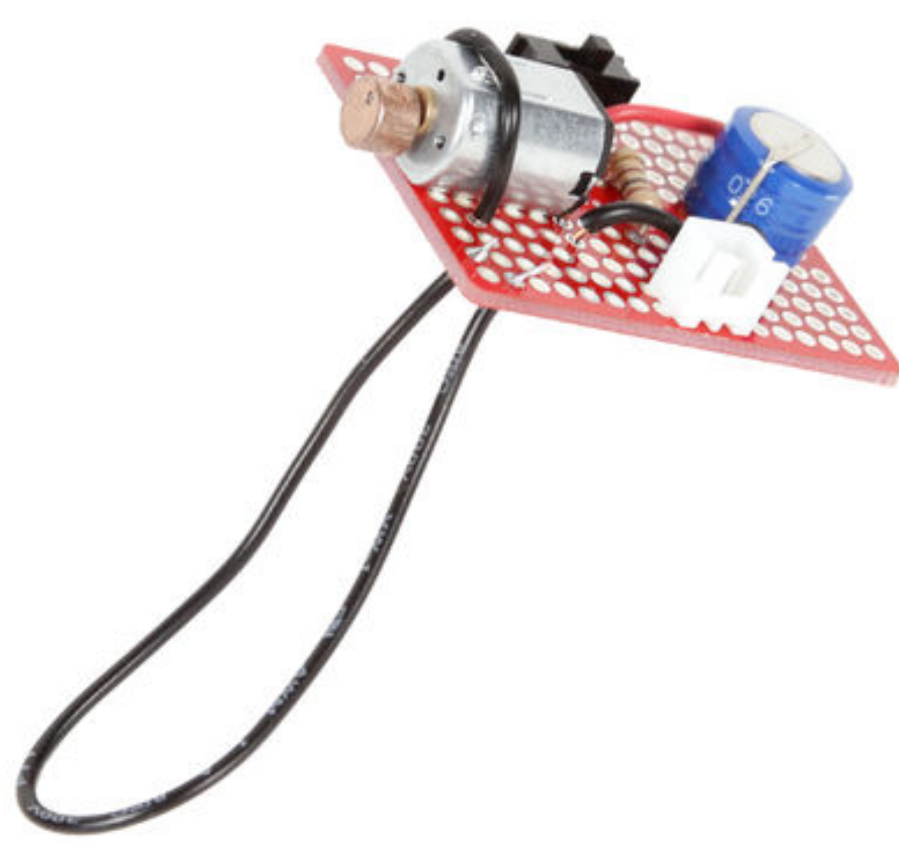
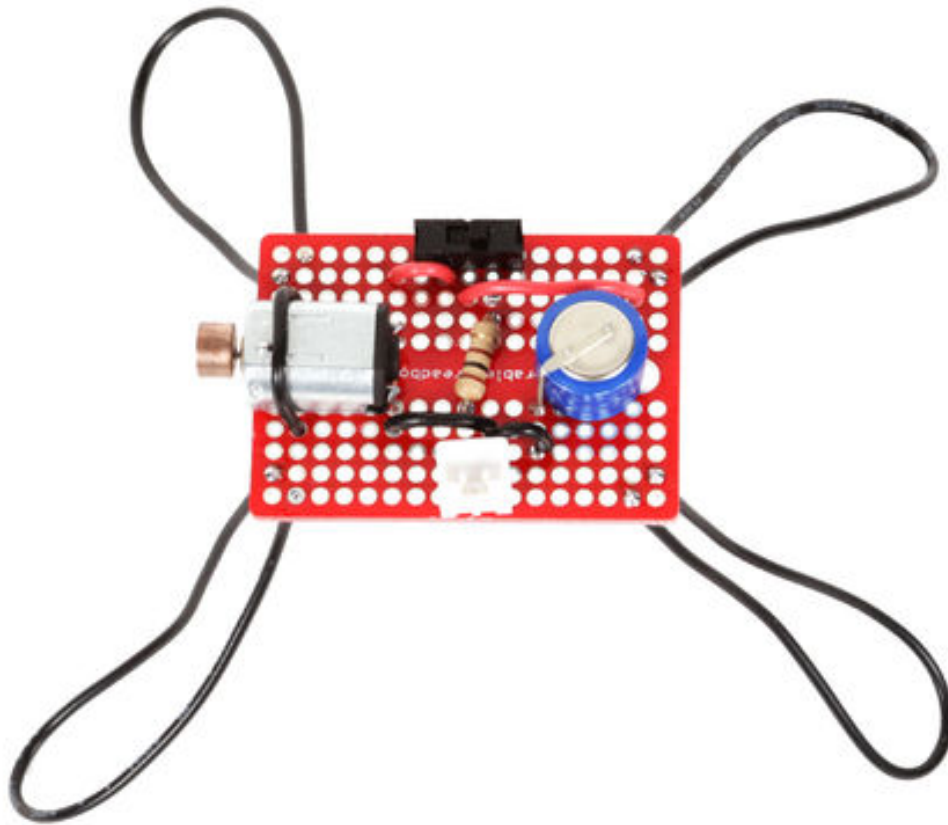
Cut four 4" solid core wires to be used as the robot's legs.

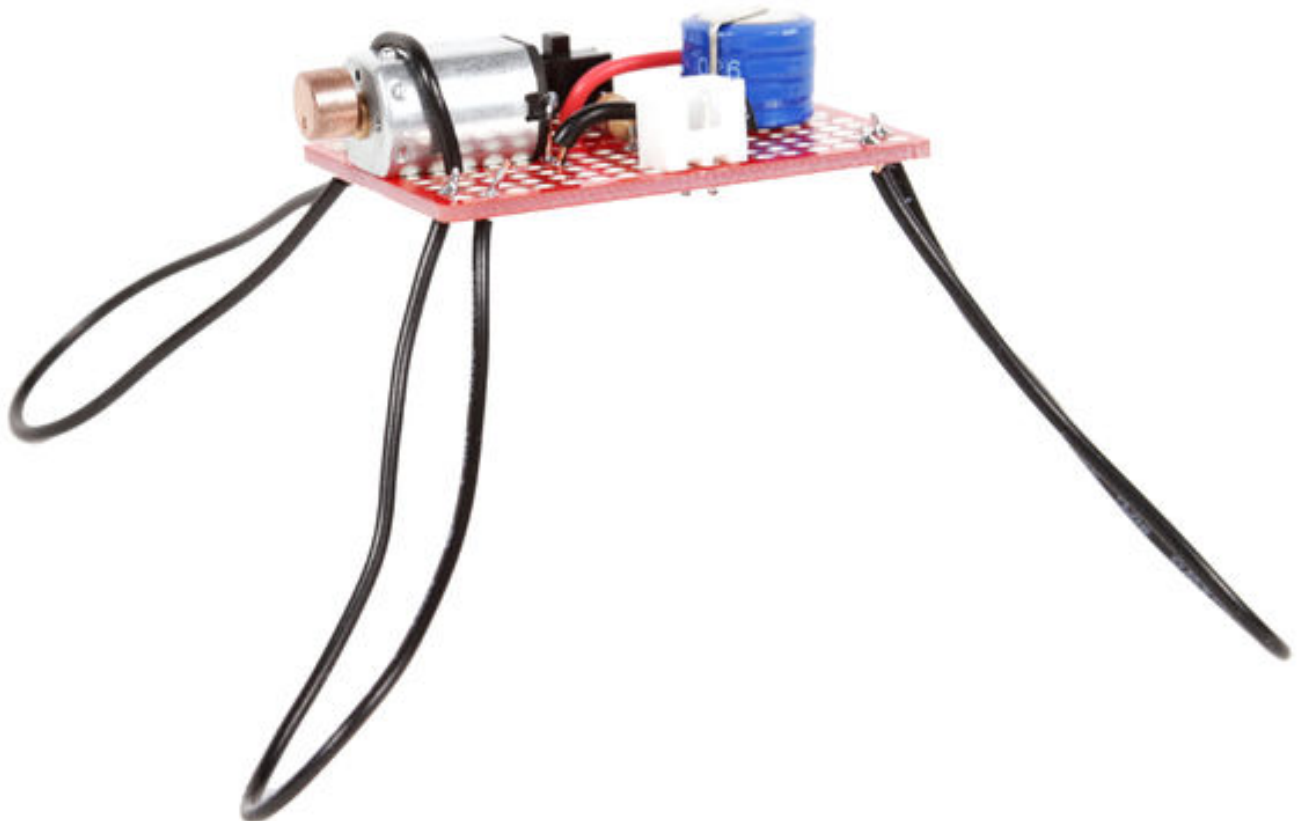


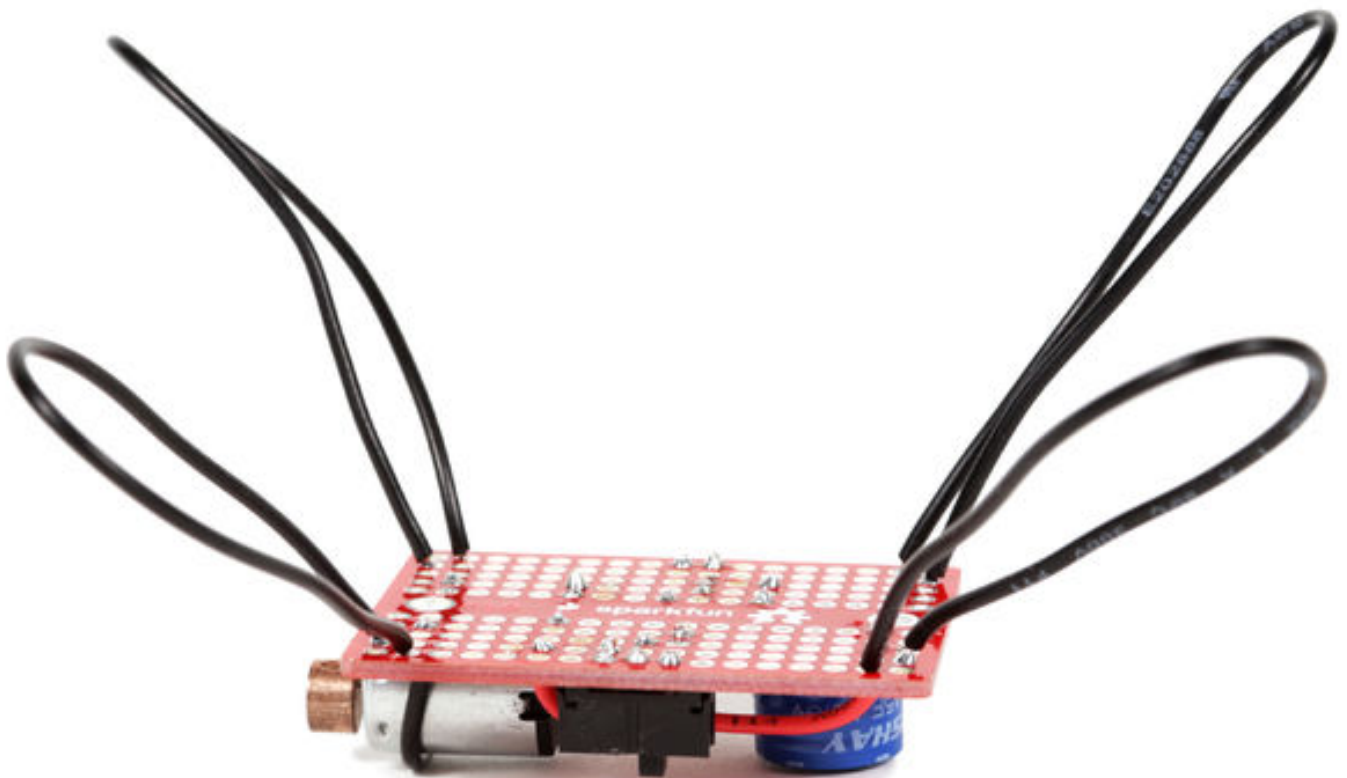
Attach the Legs

Solder both ends of each wire into the corners of the circuit board to create four wire loops.

These should not be electrically connected to any actual components on the circuit board.





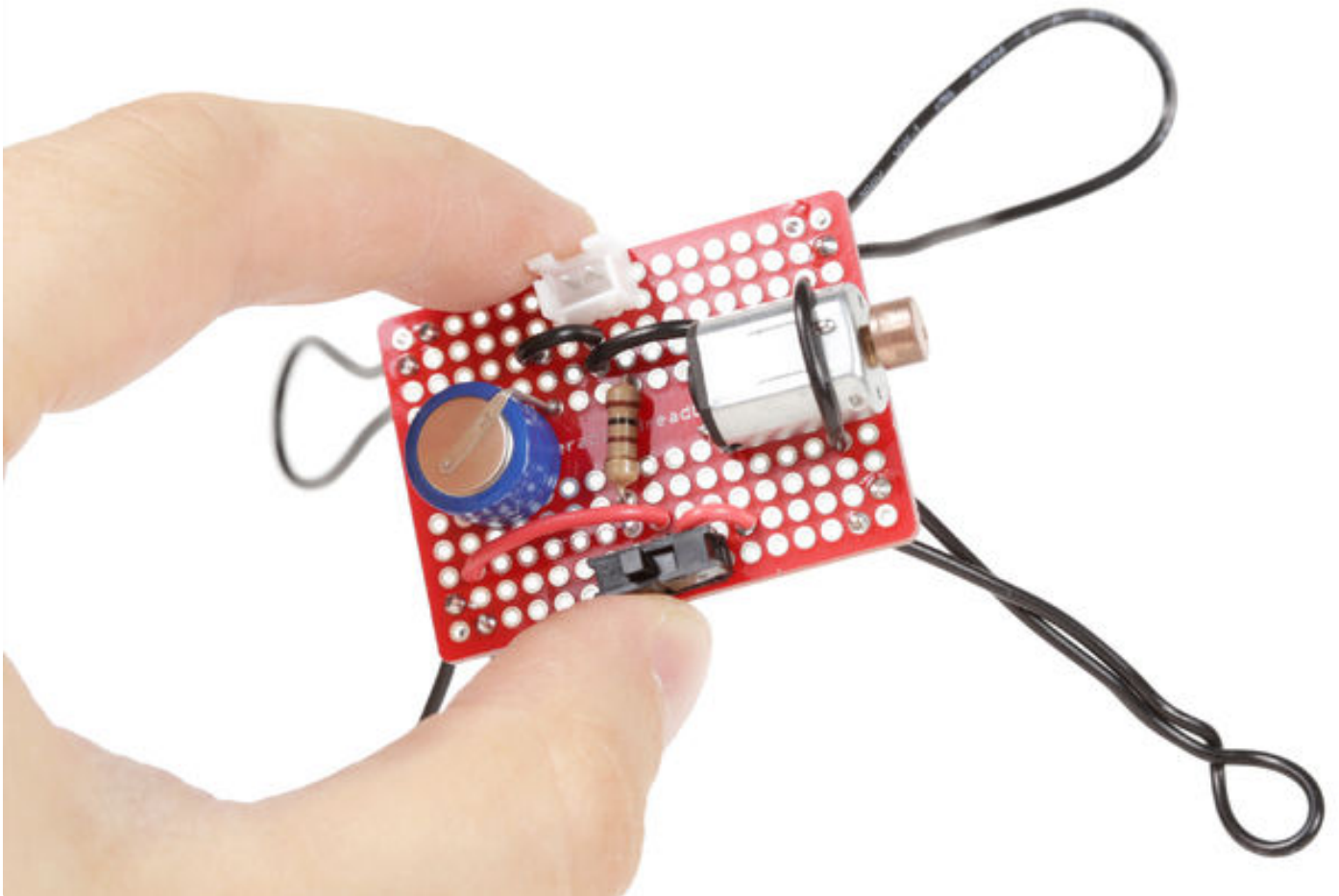
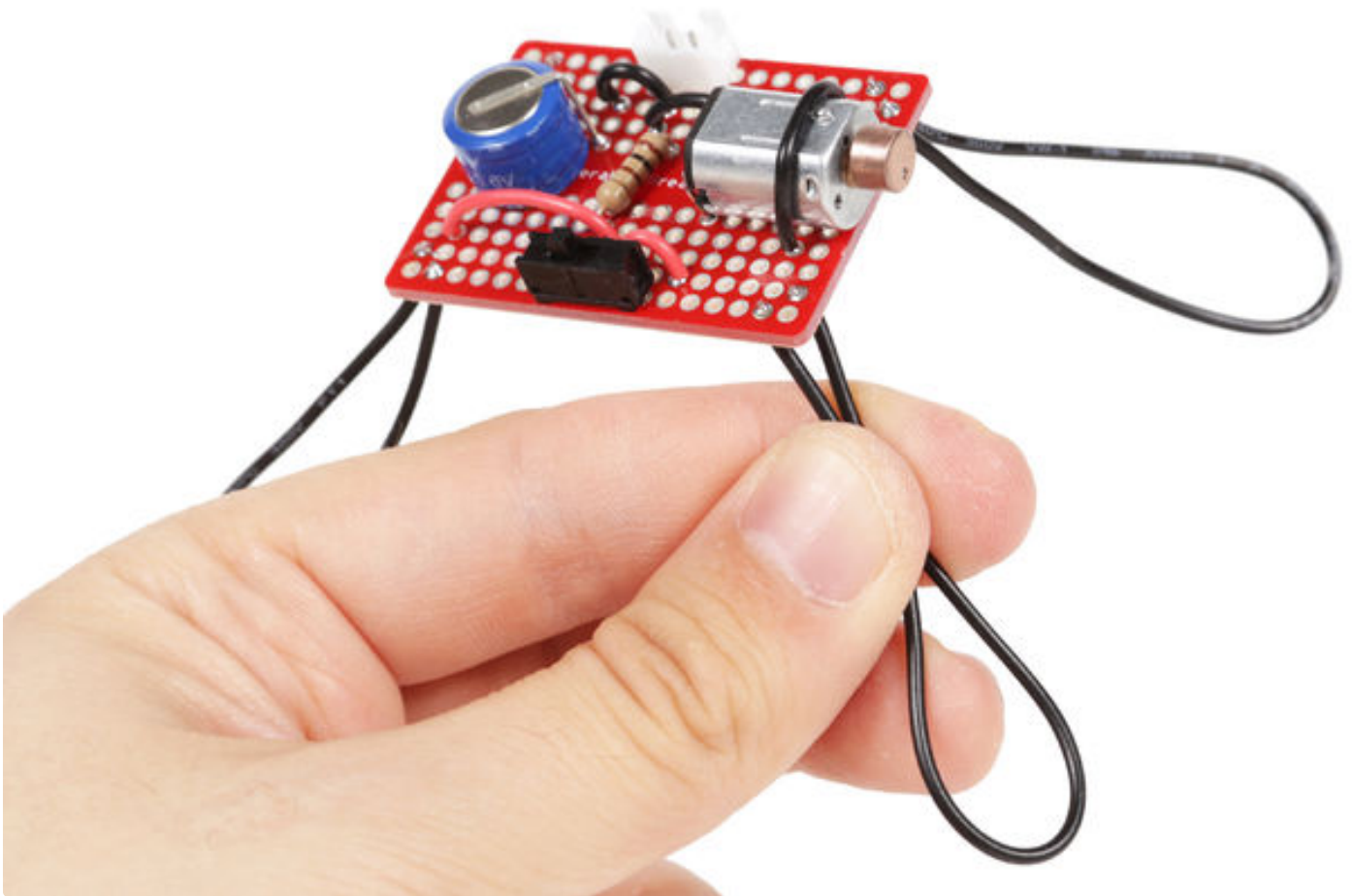


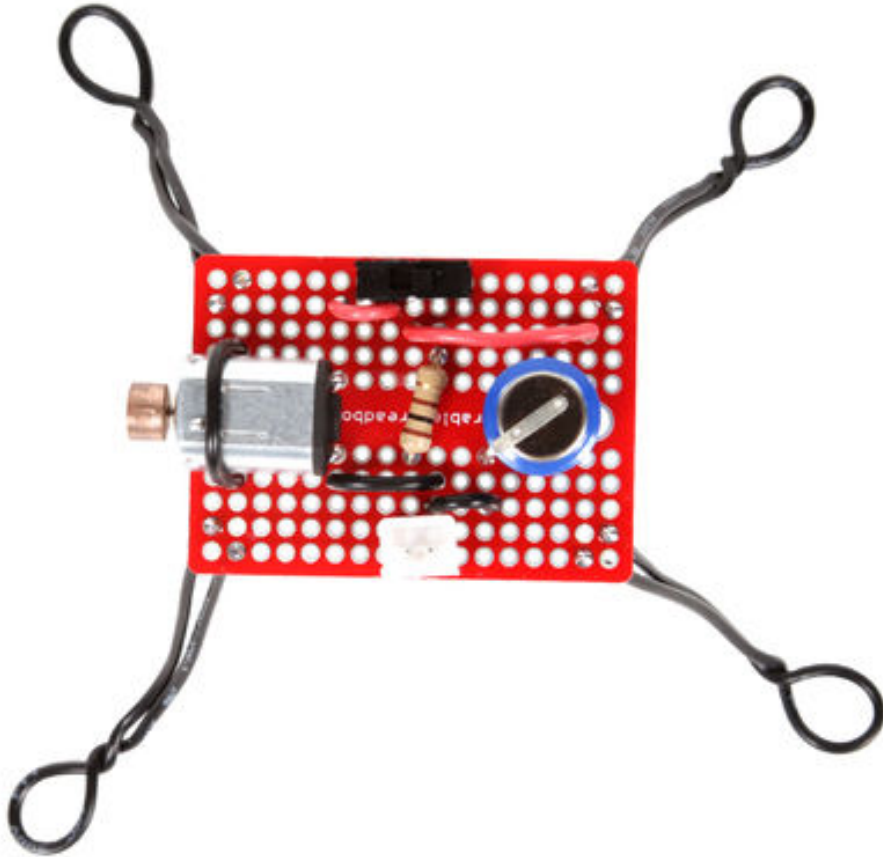
Shape the Legs

Shape all four wires into legs as you see fit.

I gave each one little loop feet, but perhaps there is another design which might work better. Feel free to experiment with form and aesthetics. There is no true right answer.







Charge It Up

To charge it up, make sure the switch is in the charging position (i.e. the motor is not running), and plug the wall wart into the socket.

You can leave it plugged into the charger as long as you want. The capacitor will stop drawing power once it's charged and be fine. Capacitors are not like batteries whose shelf-life is diminished if you leave them charging for too long without protection circuitry.



Solar

If you want to take your robot off the grid, you can add a small solar panel to charge up the capacitor when the motor is not in use.

This addition is optional.

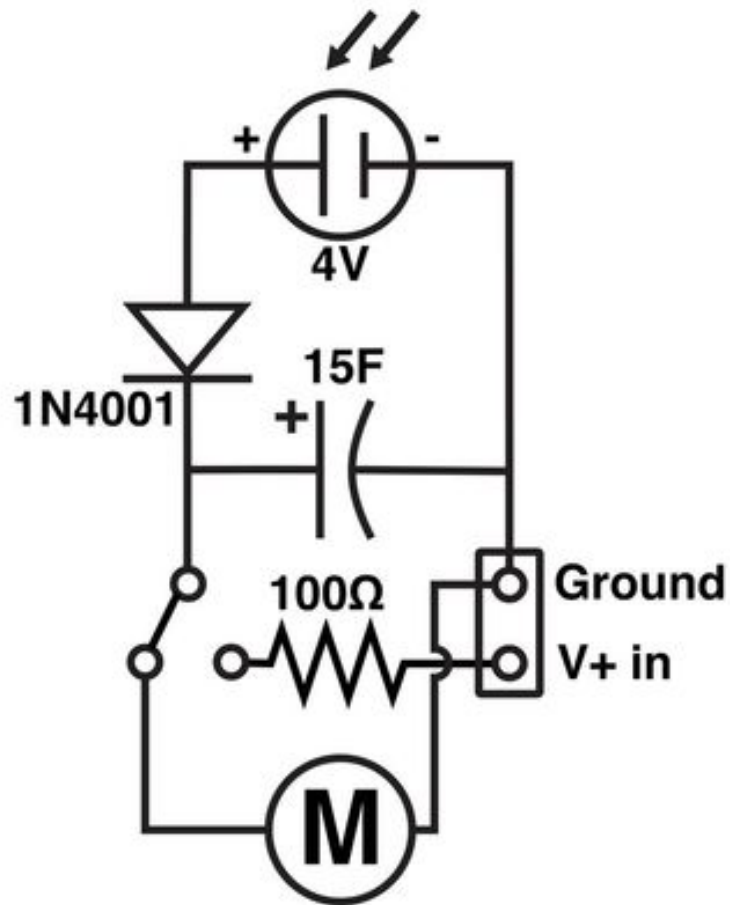


Expanding the Circuit

To make this circuit solar powered, we need to add two additional components, a solar panel and a diode.

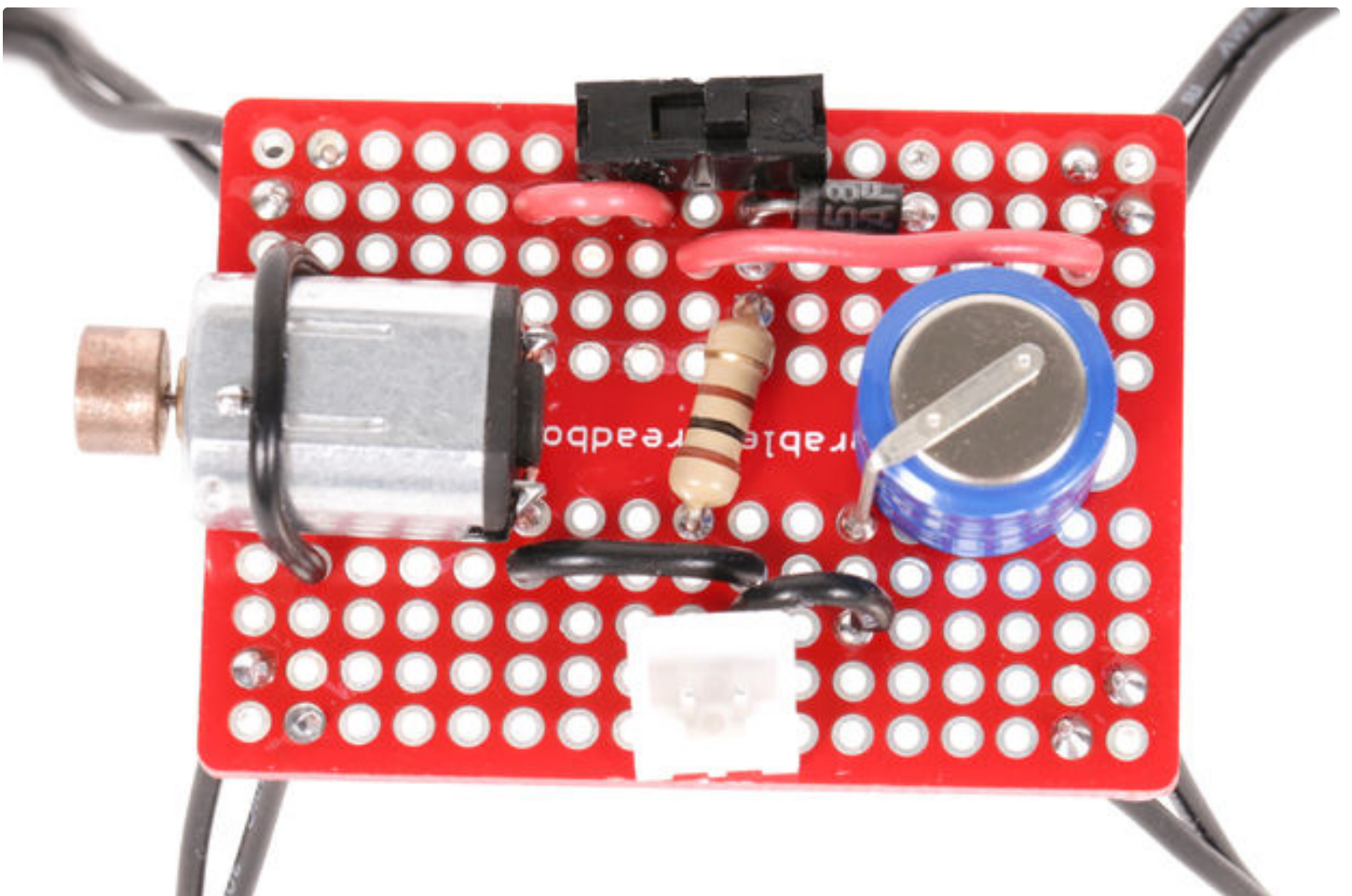
The solar panel should be rated for less voltage than the capacitor, and placed in parallel with the capacitor. Since our capacitor is rated for 5.6V, using a 4V solar panel should be safe for charging it.

We will also need to add a diode to the circuit between the positive lead on the solar panel and the capacitor. Don't yet worry too much about what diodes are. They will be discussed much further in a future lesson. For now, you just need to know that all the diode is doing is preventing electricity from the capacitor flowing backwards through the solar panel when there is no sunlight hitting it.



Adding a Diode

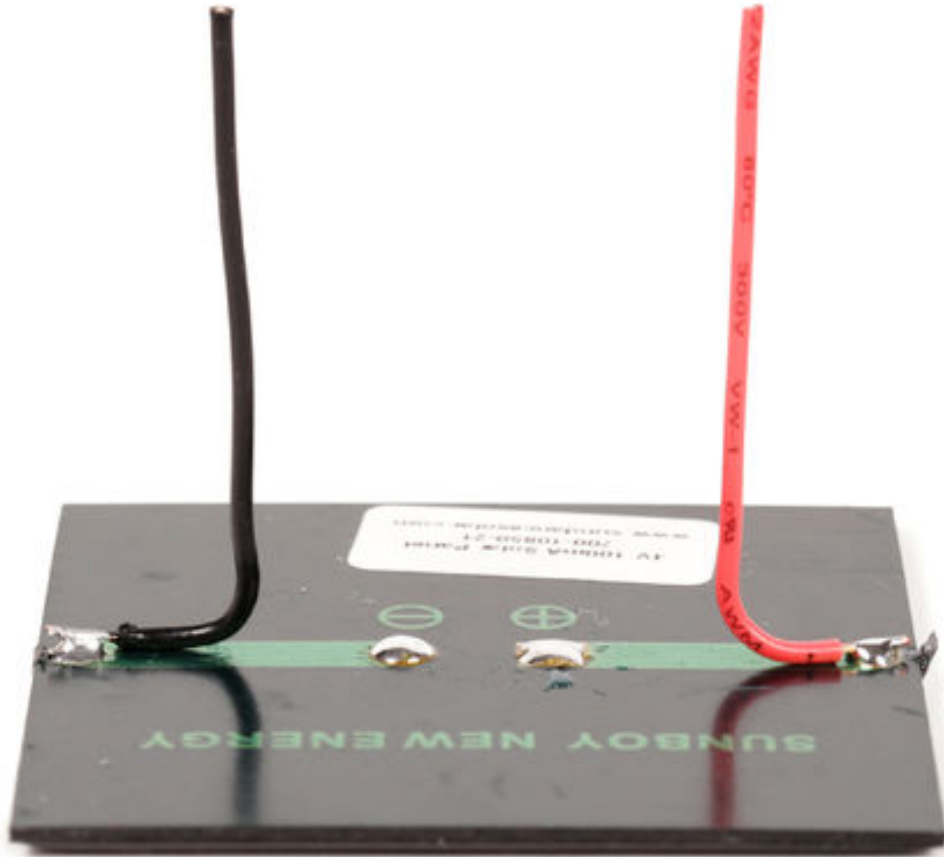
Simply connect the end of the diode with the stripe to the pin on the switch where the 100 ohm resistor is connected. Connect the other diode pin to any unused solder pad on the board.

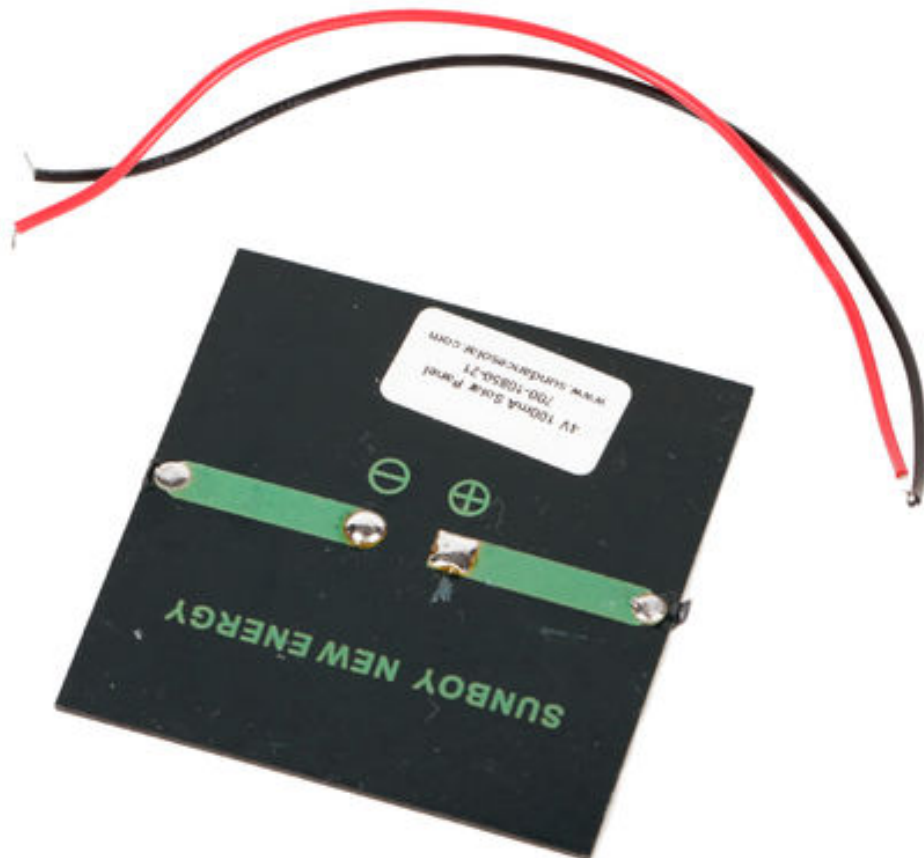


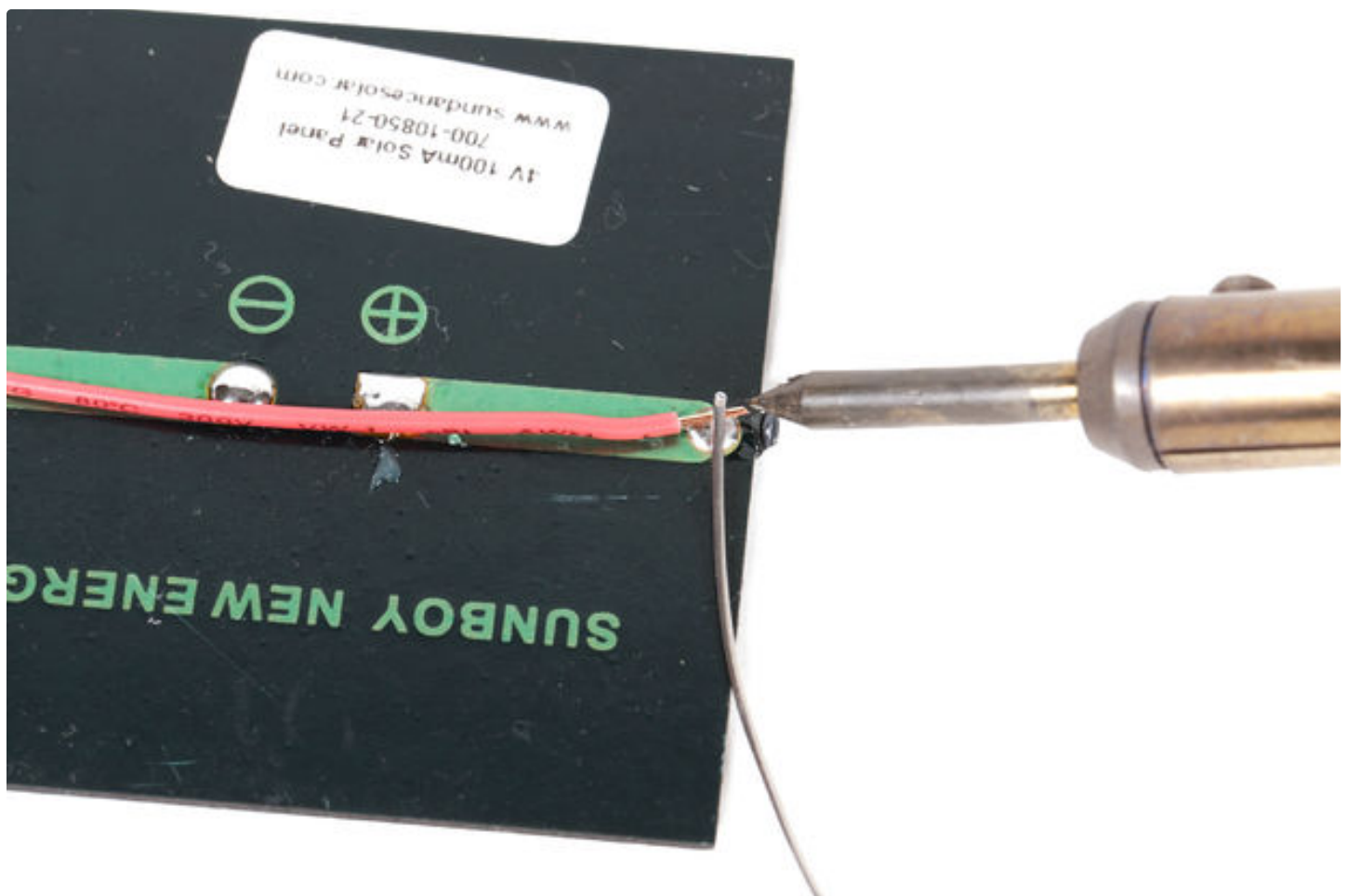
Wiring the Solar Panel

Attach a red solid core wire to the positive terminal on the solar panel and a black wire to the negative.

The reason we are replacing the existing wire with solid core wires is because these new stiffer wires will hold the solar panel in place upright above the surface of the board.







Connect the Solar Panel

Connect together the red wire from the solar panel to the unused pin on the diode.

Connect the black wire from the solar panel to any of the other ground connections on the board.

Your robot is now powered by renewable energy.

Now is time to turn your robot on and let it loose.

